

CHAPTER 15

Body Fluids and Circulation

Blood and Lymph

1. A person with blood group A Rh⁻ can receive the blood transfusion from which of the following types? (2024 Re)

A. B Rh⁻ B. AB Rh⁻ C. O Rh⁻
D. A Rh⁻ E. A Rh⁺

Choose the **correct** answer from the options given below:

- a. D and E only b. D only
c. A and B only d. C and D only

2. Which of the following statements are correct? (2023)

A. Basophils are most abundant cell of the total WBCs
B. Basophils secrete histamine, serotonin and heparin
C. Basophils are involved in inflammatory response
D. Basophils have kidney shaped nucleus
E. Basophils are agranulocyte

Choose the correct answer from the options given below:

- a. A and B only b. D and E only
c. C and E only d. B and C only

3. Match List-I with List-II. (2023-Manipur)

List-I		List-II	
(A)	Eosinophils	(I)	6 – 8%
(B)	Lymphocytes	(II)	2 – 3%
(C)	Neutrophils	(III)	20 – 25%
(D)	Monocytes	(IV)	60 – 65%

Choose the **correct** answer from the options given below:

- a. A-IV, B-I, C-II, D-III b. A-IV, B-I, C-III, D-II
c. A-II, B-III, C-IV, D-I d. A-II, B-III, C-I, D-IV

4. Arrange the following formed elements in the decreasing order of their abundance in blood in humans: (2022 Re)

A. Platelets B. Neutrophils
C. Erythrocytes D. Eosinophils
E. Monocytes

Choose the most appropriate answer from the options given below:

- a. A, C, B, D, E b. C, A, B, E, D
c. C, B, A, E, D d. D, E, B, A, C

5. Given below are two statements: (2022)

Statement-I: The coagulum is formed of network of threads called thrombins

Statement-II: Spleen is the graveyard of erythrocytes

In the light of the above statements, choose the most appropriate answer from the options given below.

- a. Statement-I is incorrect but Statement-II is correct
b. Both Statement-I and Statement-II are correct
c. Both statement-I and statement-II are incorrect
d. Statement-I is correct but Statement-II is incorrect

6. Which enzyme is responsible for the conversion of inactive fibrinogens to fibrins? (2021)

a. Renin b. Epinephrine
c. Thrombokinase d. Thrombin

7. Persons with 'AB' blood group are called as "Universal recipients". This is due to: (2021)

- a. Absence of antigens A and B in plasma
b. Presence of antibodies, anti-A and anti-B, on RBCs
c. Absence of antibodies, anti-A and anti-B, in plasma
d. Absence of antigens A and B on the surface of RBCs

8. Match the following columns and select the correct option (2020)

	Column-I		Column-II
1.	Eosinophils	(i)	Immune response
2.	Basophils	(ii)	Phagocytosis
3.	Neutrophils	(iii)	Release histaminase, destructive enzymes
4.	Lymphocytes	(iv)	Release granules containing histamine

- (1) (2) (3) (4)
a. (iv) (i) (ii) (iii)
b. (i) (ii) (iv) (iii)
c. (ii) (i) (iii) (iv)
d. (iii) (iv) (ii) (i)

9. Which of the following conditions cause erythroblastosis foetalis? (2020-Covid)

- a. Mother Rh^{-ve} and foetus Rh^{+ve}
b. Both mother and foetus Rh^{-ve}
c. Both mother and foetus Rh^{+ve}
d. Mother Rh^{+ve} and foetus Rh^{-ve}

10. Match the items given in Column-I with those in Column-II and select the correct option given below (2018)

	Column-I		Column-II
A.	Fibrinogen	i.	Osmotic balance
B.	Globulin	ii.	Blood clotting
C.	Albumin	iii.	Defence mechanism

- a. A-iii B-ii C-i
b. A-i B-ii C-iii
c. A-i B-iii C-ii
d. A-ii B-iii C-i

11. Adult human RBCs are enucleate. Which of the following statement(s) is/are most appropriate explanation for this feature? (2017-Delhi)

- A. They do not need to reproduce
B. They are somatic cells
C. They do not metabolise
D. All their internal space is available for oxygen transport
a. Only (D) b. Only (A)
c. (A), (C) and (D) d. (B) and (C)

12. Serum differs from blood in: (2016 - II)

- a. Lacking clotting factors b. Lacking antibodies
c. Lacking globulins d. Lacking albumins

13. Name the blood cells, whose reduction in number can cause clotting disorder, leading to excessive loss of blood from the body. (2016 - II)

- a. Neutrophils b. Thrombocytes
c. Erythrocytes d. Leukocytes

14. Which one of the following is correct? (2015)

- a. Lymph = Plasma + RBC + WBC
b. Blood = Plasma + RBC + WBC + Platelets
c. Plasma = Blood – Lymphocytes
d. Serum = Blood + Fibrinogen

15. Erythropoiesis starts in: (2015)

- a. Spleen b. Red bone marrow
c. Kidney d. Liver

16. If you suspect major deficiency of antibodies in a person, to which of the following would you look for confirmatory evidences? (2015 Re)

- a. Serum albumins b. Haemocytes
c. Serum globulins d. Fibrinogen in plasma

17. Person with blood group AB is considered as universal recipient because he has: (2014)

- a. Both A and B antigens in the plasma but no antibodies
b. Both A and B antigens on RBC but no antibodies in the plasma
c. Both A and B antibodies in the plasma
d. No antigen on RBC and no antibody in the plasma

18. The most abundant intracellular cation is: (2013)

- a. K^+ b. Na^+ c. Ca^{++} d. H^+

Circulatory Pathways

19. 'Lub' sound of Heart is caused by the _____. (2024 Re)

- a. closure of the semilunar valves
b. opening of tricuspid and bicuspid valves
c. opening of the semilunar valves
d. closure of the tricuspid and bicuspid valves

20. Following are the stages of pathway for conduction of an action potential through the heart: (2024)

- A. AV bundle B. Purkinje fibres
C. AV node D. Bundle branches
E. SA node

Choose the correct sequence of pathway from the options given below:

- a. B-D-E-C-A b. E-A-D-B-C
c. E-C-A-D-B d. A-E-C-B-D

21. Match List-I with List-II: (2024)

List-I		List-II	
(A)	P wave	(I)	Heart muscles are electrically silent.
(B)	QRS complex	(II)	Depolarisation of ventricles.
(C)	T wave	(III)	Depolarisation of atria.
(D)	T-P gap	(IV)	Repolarisation of ventricles.

Choose the correct answer from the options given below:

- a. A-(II), B-(III), C-(I), D-(IV)
b. A-(IV), B-(II), C-(I), D-(III)
c. A-(I), B-(III), C-(IV), D-(II)
d. A-(III), B-(II), C-(IV), D-(I)

22. Match List-I with List-II. (2023)

List-I		List-II	
(A)	P-wave	(I)	Beginning of systole
(B)	Q-wave	(II)	Repolarisation of ventricles
(C)	QRS complex	(III)	Depolarisation of atria
(D)	T-wave	(IV)	Depolarisation of ventricles

- a. A-I, B-II, C-III, D-IV b. A-III, B-I, C-IV, D-II
c. A-IV, B-III, C-II, D-I d. A-II, B-IV, C-I, D-III

23. Match List-I with List-II. (2023-Manipur)

List-I (ECG)		List-II (Electrical activity of heart)	
(A)	P-wave	(I)	Depolarisation of ventricles
(B)	QRS complex	(II)	End of systole
(C)	T wave	(III)	Depolarisation of atria
(D)	End of T wave	(IV)	Repolarisation of ventricles

Choose the **correct** answer from the options given below:

- a. A-IV, B-I, C-III, D-II b. A-I, B-IV, C-III, D-II
c. A-IV, B-III, C-I, D-II d. A-III, B-I, C-IV, D-II

24. Which one of the following statements is correct? (2022)
- Increased ventricular pressure causes closing of the semilunar valves
 - The atrio-ventricular node (AVN) generates an action potential to stimulate atrial contraction
 - The tricuspid and the bicuspid valves open due to the pressure exerted by the simultaneous contraction of the atria
 - Blood moves freely from atrium to the ventricle during joint diastole.
25. The QRS complex in a standard ECG represents: (2020)
- Depolarisation of auricles
 - Depolarisation of ventricles
 - Repolarisation of ventricles
 - Repolarisation of auricles
26. What would be the heart rate of a person if the cardiac output is 5 L, blood volume in the ventricles at the end of diastole is 100 mL and at the end of ventricular systole is 50 mL? (2019)
- 50 beats per minute
 - 75 beats per minute
 - 100 beats per minute
 - 125 beats per minute
27. Match the Column-I with Column-II (2019)

	Column-I		Column-II
A.	P - wave	i.	Depolarisation of ventricles
B.	QRS complex	ii.	Repolarisation of ventricle
C.	T - wave	iii.	Coronary ischemia
D.	Reduction in the size of T-wave	iv.	Depolarisation of atria
		v.	Repolarisation of atria

Select the correct option.

- A-iv B-i C-ii D-iii
 - A-iv B-i C-ii D-v
 - A-ii B-i C-v D-iii
 - A-ii B-iii C-v D-iv
28. Match the items given in Column-I with those in Column-II and select the correct option given below: (2018)

	Column-I		Column-II
A.	Tricuspid valve	i.	Between left atrium and left ventricle
B.	Bicuspid valve	ii.	Between right ventricle and pulmonary artery
C.	Semilunar valve	iii.	Between right atrium and right ventricle

- A-iii B-i C-ii
- A-i B-iii C-ii
- A-i B-ii C-iii
- A-ii B-i C-iii

29. Blood pressure in the mammalian aorta is maximum during: (2015)
- Systole of the left ventricle
 - Diastole of the right atrium
 - Systole of the left atrium
 - Diastole of the right ventricle
30. Which one of the following animals has two separate circulatory pathways? (2015 Re)
- Lizard
 - Whale
 - Shark
 - Frog
31. Doctors use stethoscope to hear the sounds produced during each cardiac cycle. The second sound is heard when: (2015 Re)
- Ventricular walls vibrate due to pushing in of blood from atria
 - Semilunar valves close down after the blood flows into vessels from ventricles
 - AV node receives signal from SA Node
 - AV valves open up
32. The diagram given here is the standard ECG of a normal person. The P-wave represents the: (2013)



- End of systole
- Contraction of both the atria
- Initiation of the ventricular contraction
- Beginning of the systole

Double Circulation & Regulation of Cardiac Activity

33. A unique vascular connection between the digestive tract and liver is called (2022 Re)
- Hepato-cystic system
 - Hepato-pancreatic system
 - Hepatic portal system
 - Renal portal system
34. Which of the following is associated with decrease in cardiac output? (2020-Covid)
- Parasympathetic neural signals
 - Pneumotaxic centre
 - Adrenal medullary hormones
 - Sympathetic nerves
35. The hepatic portal vein drains blood to liver from (2017-Delhi)
- Heart
 - Stomach
 - Kidneys
 - Intestine

Disorders of Circulatory System

36. Blood pressure in the pulmonary artery is: (2016 - I)
- Same as that in the aorta
 - More than that in the carotid
 - More than that in the pulmonary vein
 - Less than that in the vena cava

Answer Key

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|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (d) | 3. (c) | 4. (b) | 5. (a) | 6. (d) | 7. (c) | 8. (d) | 9. (a) | 10. (d) |
| 11. (a) | 12. (a) | 13. (b) | 14. (b) | 15. (b) | 16. (c) | 17. (b) | 18. (a) | 19. (d) | 20. (c) |
| 21. (d) | 22. (b) | 23. (d) | 24. (d) | 25. (b) | 26. (c) | 27. (a) | 28. (a) | 29. (a) | 30. (b) |
| 31. (b) | 32. (b) | 33. (c) | 34. (a) | 35. (d) | 36. (c) | | | | |

